Chia-Yi Ju（朱家誼）

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Assistant Professor

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Research Interest

Quantum Foundations/Non-Hermitian Quantum Physics, Quantum Information, String Theories, and Supersymmetric Theories.

Experience

Editorial Board Member, Physics Bimonthly (2022 -)
Assistant Professor, National Sun Yat-sen University (2022 -)
Postdoctoral Fellow, National Chung Hsing University (2016 - 2022)
Research Assistant, Tunghai University (2009 - 2010)
Lab Instructor, Tunghai University (2008 - 2009)
Assistant Squad Leader, Republic of China Marine Corps (2007 - 2008)

Education

C.N. Yang Institute for Theoretical Physics
&
Department of Physics and Astronomy,
Stony Brook University (2010 - 2016)

Ph.D. in Physics (2016)

Advisor: Warren Siegel

Department of Physics,
National Chung Hsing University (2003 - 2007)

B.Sc. in Physics (2007)

Publications

- Chia-Yi Ju, Adam Miranowicz, Yueh-Nan Chen, Guang-Yin Chen, Franco Nori, *"Emergent Parallel Transports and Curvatures in Non-Hermitian Quantum Mechanics",* Quantum **8**, 1277 (2024).
- Wan-Guan Chang, **Chia-Yi Ju**†, Guang-Yin Chen, Yueh-Nan Chen, Huan-Yu Ku, *"Visually quantifying single-qubit quantum memory",* arXiv:2312.06939 [quant-ph] (2023).
- **Chia-Yi Ju**, Wan-Yu Liu, Shuo-Yen Chen, Guang-Yin Chen, *"Prolonging the Qubits Entanglement via Local Surface Plasmon Modes",* SPIN **13**(2), 2340017 (2023).

- Chia-Yi Ju, Adam Miranowicz, Fabrizio Minganti, Chuan-Tsung Chan, Guang-Yin Chen, Franco Nori, *"Flattening the Curve with Einstein's Quantum Elevator: Hermitization of Non-Hermitian Hamiltonians via a Generalized Vielbein Formalism",* Phys. Rev. Research **4**, 023070 (2022).
- Yu-Chin Tzeng, Chia-Yi Ju, Guang-Yin Chen, Wen-Min Huang, *"Hunting for the non-Hermitian exceptional points with fidelity susceptibility",* Phy. Rev. Research **3**, 013015 (2021).
- Dharmesh Jain, Chia-Yi Ju, Warren Siegel, *"Simplifying 4d $\mathcal{N} = 3$ Harmonic Superspace",* Phys. Rev. D **102**, 066007 (2020).
- Chia-Yi Ju, Ming-Hsun Chou, Guang-Yin Chen, Yueh-Nan Chen, *"Optical quantum frequency filter based on generalized eigenstates",* Opt Express **28**, 17868 (2020).
- Chia-Yi Ju, Adam Miranowicz, Guang-Yin Chen, Franco Nori, *"Non-Hermitian Hamiltonians and no-go theorems in quantum information",* Phys. Rev. A **100**, 062118 (2019).
- Chia-Yi Ju, Warren Siegel, *"Gauging Unbroken Symmetries in F-theory",* Phys. Rev. D **94**, 106004 (2016).
- Chia-Yi Ju, Warren Siegel, *"Systematizing semishortening conditions",* Phys. Rev. D **90**, 125004 (2014).
- Fei Fang Chung, Sy-Sang Liaw, Chia-Yi Ju, *"Brazil nut effect in a rectangular plate under horizontal vibration",* Granul. Matter **11**(2), 79 (2009).
- Fei Fang Chung, Chia-Yi Ju, Sy-Sang Liaw, *"Spiral trajectory in the horizontal Brazil nut effect",* Phys. Rev. E **77**(6), 061304 (2008).

Talks

Conferences/Workshops

- International Workshop on Quantum Information Processing and Applications 2022 (Invited Talk), *"Something Will Come of Something — Emergent Dimensions"*, Dec. 11, 2022.
- 2022 Workshop on Quantum Science and Technology (Invited Talk), *"Roaming in the Parameter Space: Emergent Dimensions"*, Aug. 26, 2022.
- Pseudo-Hermitian Hamiltonians in Quantum Physic XIX<vPHHQP<XX (Invited Talk), *"Einstein's Quantum Elevator: Hermitization of Non-Hermitian Hamiltonians via a Generalized Vielbein Formalism"*, Jul. 7, 2022.
- 2022 Young Researchers Forum on Quantum Information Science (Invited Talk), *"The Geometry of Quantum States"*, Feb. 11, 2022.
- 2021 Non-Hermitian Physics (Online) ICTS, Bengaluru, India, *"The Geometry of Quantum States"*, Mar. 24, 2021.
- 2020 AMO Summer School (Invited Talk), *"The Dynamics and Deformations of Hilbert Spaces"*, Sep. 1, 2020.
- Workshop NH2019TD | IIS-Chiba | The University of Tokyo (Invited Talk), *"The Geometries of Quantum States and the No-Go Theorems"*, Nov. 28, 2019.
- Workshop NH2019 | IIS-Chiba | The University of Tokyo (Invited Talk), *"From Geometry of NHQM to No-Go Theorems"*, Aug. 8, 2019.
- 2018 Workshop on Quantum Science and Technology, *"A Possible Quantum Filter"*, Sep. 4, 2018.

- 2018 Young Researchers Forum on Quantum Information Science, *"Quantum Resonanators"*, Jun. 30, 2018.

Seminars/Colloquiums

- National Taiwan Normal University, *"Emergent dimensions, evolutions, and event-horizon-like phenomena"*, Oct. 25, 2023.
- RIKEN, *"How hidden dimensions can emerge naturally from parameter spaces and derive the evolution equations for quantum states"*, Jun. 27, 2023.
- National Yang Ming Chiao Tung University, *"Emergent Dimensions from the Parameter Space"*, May 30, 2023.
- Chung Yuan Christian University, *"Something Will Come of Something — Emergent Dimensions"*, Dec. 21, 2022.
- National Sun Yat-sen University, *"An Informal Introduction to Formal Theory – Non-Hermitian Quantum System Inspired Formalism"*, Nov 10, 2022.
- National Cheng Kung University, *"Roaming in the Parameter Space – Emergent Dimensions"*, Oct 31, 2022.
- National Sun Yat-sen University, *"An Informal Introduction to Formal Theory – From Geometry to Quantum Physics"*, Mar 31, 2022.
- National Sun Yat-sen University, *"The Geometry of Quantum States"*, Sep 30, 2021.
- National Cheng Kung University, *"The Geometry of Quantum States"*, May 6, 2021.
- National Cheng Kung University, *"A Geometrical Phase Transition"*, Mar 22, 2021.
- National Cheng Kung University, *"The Dynamics and Deformations of Hilbert Spaces"*, May 11, 2020.
- Tunghai University, *"The Geometry of (Non-)Hermitian Quantum Mechanics"*, Mar 13, 2020.
- National Taiwan University, *"A Geometric Approach to Quantum Mechanics"*, Jun 14, 2019.
- National Cheng Kung University, *"A Geometric Approach to Quantum Mechanics"*, Apr 8, 2019.
- National Changhua University of Education, *"Geometric Approach to Quantum Mechanics"*, May 1, 2019.
- National National Tsing Hua University, *"A Possible Quantum Filter"*, Oct 23, 2018.
- National Kaohsiung Normal University, *"A Possible Quantum Filter"*, Oct 17, 2018.
- National Cheng Kung University, *"A Possible Quantum Filter"*, Sep 17, 2018.
- National Kaohsiung Normal University, *"To String, or Not to String – Beyond Standard Model"*, Apr 11, 2018.
- National Chung Hsing University, *"To String, or Not to String – Beyond Standard Model"*, Jan 12, 2018.